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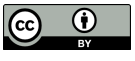
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Hybrid Deep Learning Models for Remote Sensing Image Processing

Matthieu Muller, Daniele Picone, Begüm Demir, Gustau Camps-Valls, Mauro Dalla Mura, Magnús Örn Úlfarsson, Jón Atli Benediktsson

***Abstract*—Core image processing tasks, such as super-resolution, denoising, deblurring, pansharpening, and atmospheric correction, underpin all optical Remote Sensing (RS) pipelines. Errors at this stage propagate through downstream applications, distorting land-cover maps, change detection, and climate records. Classical physics-based models capture sensor optics, radiometry, and geometry but struggle with complex noise and scene variability. In contrast, Deep Learning (DL) methods offer powerful data-driven solutions yet often act as black boxes, ignoring physical constraints and overfitting to spurious patterns. Hybrid Deep Learning (HDL) approaches bridge this gap by integrating physical models with neural architectures, combining interpretability and data adaptivity. This review surveys the emerging landscape of HDL methods in RS image processing, outlining their theoretical foundations, motivations, and design philosophies. We categorize fusion strategies, from model-embedded schemes (e.g., Plug-and-Play, Unrolling) to model-guided learning (e.g., Deep Image Prior, unsupervised frameworks), and discuss how they enhance trust, robustness, and physical consistency in RS image analysis.**

***Index Terms*—Remote Sensing, Image Processing, Plug and Play, Unrolling, Deep Image Prior, Deep Learning, Physical Model, Hybrid Method**

I. Introduction

Remote Sensing (RS) is the science of obtaining information about objects or processes without direct physical contact. By exploiting the interaction between electromagnetic radiation and matter, instruments mounted on satellites, aircraft, or unmanned aerial vehicles capture multi-modal observations that can span the entire globe, revisit the exact location within hours, and penetrate otherwise inaccessible or hazardous environments. These large-scale and repeatable measurements have become indispensable for monitoring Earth system dynamics, assessing natural resources, and supporting precision agriculture. Moreover, remote sensing plays a vital role in understanding and monitoring Earth's climate change. It helps to guide disaster preparation and response, enabling climate change mitigation and adaptation strategies. Thanks to continuous advances in sensor engineering, data infrastructures, and algorithmic processing, remote sensing has become a critical pillar of modern science, policy, and industry, providing valuable insights at spatial scales ranging from individual buildings to the entire planet.

The choice of sensors is crucial for obtaining accurate, meaningful, and application-specific measurements in remote sensing. Optical sensors, like the European Space Agency's Sentinel-2 [1] and the U.S. Geological Survey's Landsat program [2], are among the most widely used due to their ability to capture detailed surface information through the detection of solar radiation that is reflected or emitted by the Earth's surface. These sensors operate across a varied range of wavelengths, including visible, near-infrared, and shortwave infrared, and enable the monitoring of various land characteristics such as vegetation [3], [4], surface temperature [5], and urban development [6].

Multispectral sensors collect data across a limited number of broad spectral bands and are effective for general land-cover monitoring. On the other hand, hyperspectral sensors such as AVIRIS [7] and ENMAP [8] acquire data across hundreds of narrow spectral bands, providing rich spectral signatures that enable precise material identification and sub-pixel classification. However, the performance of optical sensors can be compromised by atmospheric conditions, particularly cloud cover and varying illumination, which can obscure or distort the observed data.

A. Image Processing Tasks in Remote Sensing

The remote sensing field encompasses a diverse set of tasks aimed at extracting meaningful information about the Earth's surface and atmosphere from data acquired by different sensors. While high-level tasks such as land cover classification and climate/weather forecasting are often highlighted, image processing tasks form the foundation for all accurate and interpretable analysis. These tasks are crucial for enhancing the quality of raw data and ensuring reliable outcomes in downstream applications.

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Below is an overview of the main image processing tasks in remote sensing, illustrating the wide range of challenges involved in achieving more faithful Earth imagery and ultimately, a more comprehensive understanding of our planet.

a) Denoising: Seeks to enhance degraded images by suppressing noise artifacts originating from multiple sources. By restoring clarity, these methods improve interpretability, especially in challenging conditions such as low illumination. Denoised images contribute to more reliable conclusions in downstream tasks such as land cover classification [9]–[11], change detection [12], and anomaly detection [13].

b) Deblurring: Focuses on image degradation caused by motion, atmospheric turbulence, or limitations in sensor optics, and tries to recover sharp spatial structures. Accurate deblurring enhances spatial resolution [14] and enables finer analysis of texture and edges, particularly useful for image segmentation [15].

c) Single Image Super-Resolution: In this task, low-resolution observations are transformed to generate high-resolution outputs, improving clarity and revealing finer spatial details that might be lost in the original data. Super-resolution can benefit many downstream applications, such as segmentation or land cover classification [16], where sharper imagery helps identify subtle features like narrow roads or small structures [17].

d) Sharpening: Remote sensing often involves collecting images at various spatial resolutions or from different sensor modalities, such as other satellites. The goal of sharpening is to integrate multi-resolution imagery by upscaling lower-resolution bands to match the highest spatial resolution. This approach has been effectively applied to Sentinel-2 data, as shown in [18], [19]. The paper [20] demonstrates the value of Sentinel-2 sharpening for water-body mapping. Other examples include multi/hyperspectral fusion [21], and Pansharpening [22]–[24], which integrates a high-resolution panchromatic band with lower-resolution multispectral or hyperspectral observations to deliver images with both high spatial and spectral resolution, with direct agricultural applications [25].

e) Hyperspectral Unmixing: Hyperspectral sensors capture hundreds of narrow spectral bands, with each pixel containing a mixture of spectral signatures from multiple materials. Unmixing breaks down these mixed signatures to identify the present materials, known as endmembers, along with their relative abundances [26]. This granular view of surface composition aids classification tasks [27] and supports applications such as identifying mineral deposits [28], [29], monitoring soil properties [30], and assessing conditions on other celestial bodies, including Mars [31].

f) Compression: RS sensors usually acquire a large amount of data that cannot be directly sent to the ground stations due to limited bandwidth. Image compression reduces the size of data to be transmitted back to Earth for further processing and analysis [32]. Once received, the data are decompressed, possibly resulting in information loss if the compression method is "lossy".

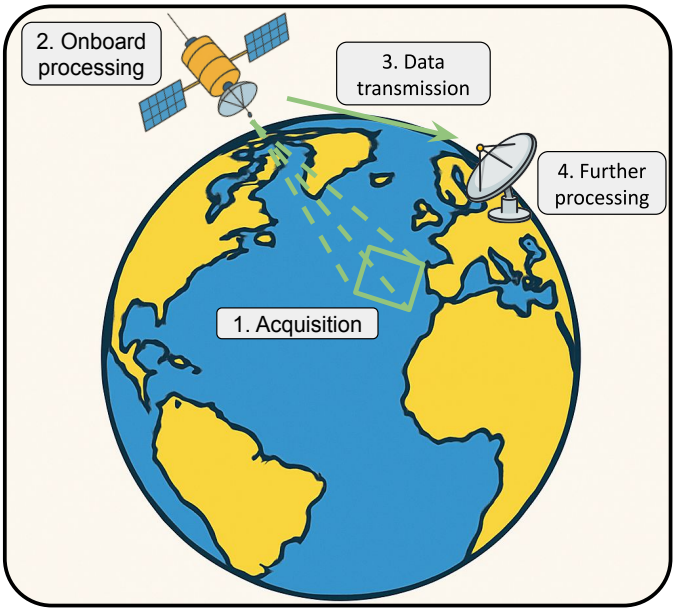


Fig. 1: Schematic of an acquisition pipeline. The data are first captured by the sensor, processed, and sent to the ground station for further refinements.

g) Demosaicking: Some snapshot imagers [33], [34] capture data through band-filter mosaics, acquiring only one spectral sample per pixel location. Demosaicking consists of recovering the missing channels by exploiting spatial-spectral correlations and sensor models. Performed either on board or at a ground station, demosaicking interpolates the incomplete mosaic into a complete multispectral cube [35], for example, allowing monitoring of Earth's aurora [36].

Fig. 1 illustrates the usual remote sensing acquisition process. The sensor captures the data from the desired scene. Due to limited onboard computational power, the data are compressed and sent to the ground station, where all the necessary processing is conducted.

B. Evolution of Methods in Remote Sensing

The methodological landscape of the RS field has undergone significant evolution over the past few decades, reflecting advances in computational capabilities, data availability, and machine learning theory.

a) Model-Based Methods: Historically, remote sensing analysis has been dominated by model-based methods that rely on physical principles and analytical formulations to extract meaningful information from sensor data. These methods explicitly encode knowledge of the sensing process, atmospheric effects, and surface interactions through radiative transfer models, inverse problem formulations, and handcrafted regularization techniques. For example, atmospheric correction algorithms use detailed physics to estimate surface reflectance from top-of-atmosphere radiance [37] At the same time,

spectral unmixing models decompose hyperspectral signatures using linear or nonlinear mixing assumptions [38].

These physically grounded models offer interpretability and robustness but are often limited by their simplifying assumptions, computational complexity, and dependence on expert-defined parameters. They also struggle to adapt to diverse environmental conditions or sensor types.

b) The Rise of Deep Learning and Artificial Intelligence: With the increasing availability of large-scale, labeled datasets and the growing accessibility of high-performance computing, particularly GPUs, the field has seen a dramatic shift towards deep learning-based methods [39]. Accessible and open-source frameworks, such as PyTorch [40], TensorFlow [41], and JAX [42], allow researchers to build and experiment with deep neural networks with ease. Nowadays, Convolutional Neural Networks (CNNs) [43], and transformer architectures [44], [45] often outperform traditional models in classical image processing tasks such as denoising, sharpening, and unmixing.

Deep Learning (DL) methods excel at automatically learning hierarchical representations from raw data, offering improved performance in complex, high-dimensional settings. They are particularly effective for integrating multimodal sources, detecting subtle patterns, and scaling to global archives. However, these models often operate as black boxes, lack physical interpretability, and require vast amounts of annotated data for training. Their performance also degrades when exposed to distribution shifts across regions, sensors, or seasons.

c) Toward Hybrid Methods: In recent years, there has been a growing shift towards the development of Hybrid Deep Learning (HDL) methods that combine the strengths of physics-based modeling and data-driven learning. These methods aim to improve generalization, interpretability, and robustness by embedding physical knowledge into learning-based architectures [46]. Hybrid methods offer a middle ground: They retain the domain knowledge and trustworthiness of physical models while leveraging the adaptability and expressiveness of machine learning [47].

Rather than training a neural network to learn every aspect of a task from raw data, which often requires large datasets and computational resources, hybrid models introduce structure and prior knowledge through physical modeling. This reduces the learning burden as the network no longer needs to discover well-understood principles, such as sensor response, from scratch. Instead, it can focus on learning complex residual patterns, modeling noise characteristics, or adapting to real-world deviations from theoretical assumptions.

These hybrid methods can be classified into two categories, depending on how the physical models are used. First are the *Deep Learning Assisted Methods*, where the physical model actively participates in the method itself, typically conditioning model-specific operations. This category encompasses well-known methods such as Plug-and-Play (PnP) [46] and algorithm unrolling [48]. Second are the *Model-Guided Deep Learning Methods*, which are more focused on the DL framework and rely on physical models solely during training time. Such Model-Guided methods are Deep Image Prior (DIP) [49] and other unsupervised methods.

As the RS field continues to generate increasingly diverse and voluminous data, the convergence of model-based reasoning and data-driven learning is expected to play a central role in the development of next-generation geospatial analytics. Fig. 2 highlights the different classes of hybrid methods that fill the gap between fully model-based and end-to-end deep learning methods.

II. A GENERIC FRAMEWORK FOR RS IMAGE PROCESSING AND ANALYSIS

A. Defining "Physics" and "Model" in Remote Sensing

a) Physics: In the context of remote sensing, we define *physics* as all the fundamental principles, from nature's behavior to the sensing processes. Most notably, *physics* corresponds to:

- the laws that govern natural processes, like the interactions between electromagnetic waves, Earth's surface, and the atmosphere;
- the theoretical principles upon which the sensing techniques are based.

This encompasses well-established theories like electromagnetic wave propagation, radiative transfer, dynamics of the atmosphere and land surface. It also includes the optical concepts and information theories that occur during the acquisition process.

b) Physical Model: We define a *physical model* as any representation or implementation of the physical laws described previously. Such models allow linking together data in a practical and plausible way, respecting physical principles found in nature. They often take the form of mathematical expressions (such as partial differential equations, stochastic processes, or algorithms) or numerical simulators.

Because the underlying physical laws are often complex and only partially known, a *physical model* is always an approximation of the physical concepts it represents. It is essential to keep this in mind while exploring model-based methods, as the needed precision and available resources will determine what is an "acceptable" degree of approximation.

c) Sensing Model: More specifically, we define a *sensing model* as a physical model that represents the laws that link the actual natural scene to the observation acquired by a specific sensor. It accounts for the sensor's characteristics, such as spectral responses, spatial resolution, optical system, and sensitivity, to predict how the sensor will observe the physical scene under different conditions.

B. Inverse Problem Formulation: A Unifying Framework

In remote sensing, the acquisition process leading to the observed data is often (partially) known. The specifications of the imaging systems, including their calibration, resolution, and sensor type, are typically available through technical documentation and product specifications. For example, the Sentinel-2 mission, developed by the European Space Agency (ESA),

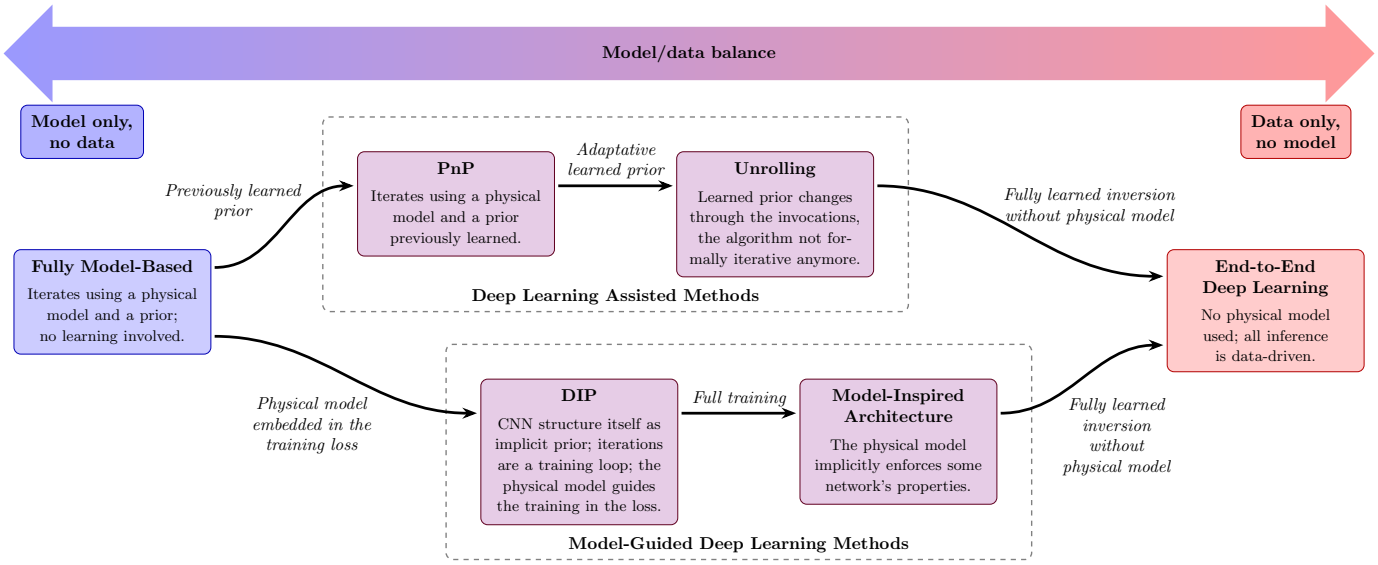


Fig. 2: Overview of the inference for the main reviewed methods in the paper. At one end, fully model-based methods, presented in Section III-A, only use physical knowledge without relying on the AI framework. At the other end, end-to-end deep learning, presented in Section III-B, only uses training data without any prior knowledge. *Deep Learning Assisted Methods*, presented in Section IV-A, utilize *physical models* directly within the algorithms, maintaining a strong connection with model-based methods. *Model-Guided Deep Learning Methods*, presented in Section IV-B, embed *physical models* in the training phase, drawing more inspiration from DL methods.

provides detailed information about its multispectral imaging system, including the spectral bands, spatial resolution, and radiometric properties of the sensor [1]. Similarly, the Landsat program, operated by NASA and the U.S. Geological Survey (USGS), offers the characteristics of the Landsat 8 Operational Land Imager (OLI), which provides data on spectral bands and spatial resolution [2].

Typically, this knowledge can be cast as a *sensing model*, also called a forward model, linking the scene and its observation. The *sensing model* may take diverse formulations depending on the available sensors and the kind of processing applied to the observations. For example, in the case of:

- Time-series observations, the *sensing model* captures the sensor's specifications and a discrete-time evolution operator [50], [51];
- Multispectral or hyperspectral data cubes, for which the *sensing model* encodes the band-specific spectral response;
- Multiple sensors' observations, where the *sensing model* represents the sensors along with the fusion process [52].

The *sensing model* translates mathematically into an (possibly non-linear) operator $\boldsymbol{A}$ and the observation process is described as:

$$\boldsymbol{y} = \boldsymbol{A}(\boldsymbol{x}) + \boldsymbol{\Gamma}(\boldsymbol{x}) + \boldsymbol{\epsilon},$$

where $\boldsymbol{x} \in \mathbb{R}^m$ is the unknown vectorized image to be estimated (for example, surface reflectance or material composition) in a vector shape, $\boldsymbol{y} \in \mathbb{R}^n$ is the vector observation of $\boldsymbol{x}$ following the *sensing model* $\boldsymbol{A}$. $\boldsymbol{\Gamma}(\boldsymbol{x}) \in \mathbb{R}^n$ represents a signal-dependent noise [53], [54] such as shot noise. Additionally, the vector $\boldsymbol{\epsilon} \in \mathbb{R}^n$ is a signal-independent noise vector, usually Gaussian.

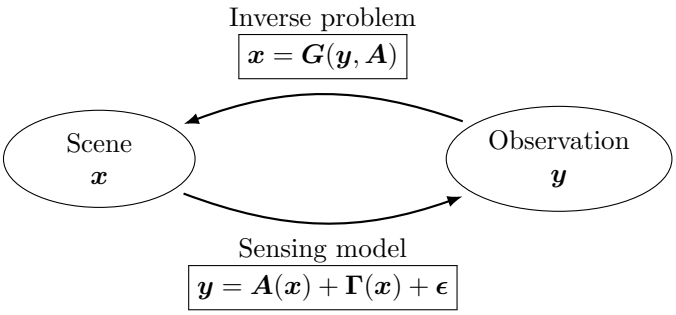


Fig. 3: The inverse problem is the counterpart of the sensing model: Finding $\boldsymbol{x}$ using $\boldsymbol{y}$.

Given a specific *sensing model*, one can define a so-called inverse problem, where the goal is to define an operator $\boldsymbol{G}$, able to recover the scene $\boldsymbol{x}$ from observed measurements $\boldsymbol{y}$. Fig. 3 illustrates the relationship between the *sensing model* and the inverse problem.

However, despite the availability of such specifications, it may not always be sufficient to recover the original data from the observed measurements fully. For instance, the imaging system may be partially damaged or not functioning optimally, leading to uncertainties in the observed data. In some cases, only partial or approximate information about the system's parameters may be accessible due to either calibration uncertainties or a lack of documentation. Additionally, uncertainties in the external environment, such as atmospheric conditions and cloud cover, can further complicate the interpretation of the observed scene. For example, in airborne or spaceborne Earth observation, clouds may obscure the land surface, pre-

venting direct observation of specific areas and impacting the reconstruction process. Even if no such problems are encountered, when the *sensing model* is linear, it is usually not a square matrix and is rank deficient, that is, $n < m$, making the problem ill-posed. In that setting, it cannot be expected to invert the *sensing model* and exactly recover the vector $\boldsymbol{x}$ using only $\boldsymbol{y}$ and $\boldsymbol{A}$.

In such instances, where knowledge of the acquisition system does not suffice for complete reconstruction, prior knowledge of the observed scene will enforce expected structures in the reconstruction independently of the acquisition system. This prior knowledge, distinct from the system's specifications, refers to information about the *physics* of the scene itself and its inherent properties. It can encompass a wide range of characteristics, including expected structures (e.g., urban areas, vegetation, water bodies), typical spectral responses (e.g., the spectral signatures of different land cover types), and known physical or geological constraints (e.g., expected terrain slopes or vegetation types). This information is derived from *physical models*, such as historical data or physical laws that govern the observed phenomena. By incorporating such prior knowledge, the reconstruction process can be guided to yield more accurate results, even when faced with missing or uncertain data from the imaging system.

The following sections present the different methods to solve such inverse problems in remote sensing, spanning classical model-based methods, newer end-to-end deep learning in Section III, and HDL methods mixing model-based theory and deep learning in Section IV, as summarized by Fig. 2. We will see that with the emergence of deep learning methods, a paradigm shift has happened. Indeed, while model-based methods primarily rely on *physical models* to design algorithms, deep learning methods emphasize the algorithm itself, abstracting the underlying *physics*. Table I summarizes the strengths and limitations of all three families of RS methods considered in this work.

III. Classical methods

A. Fully Model-based Methods

The most common way to effectively make use of the available *physical models*, including *sensing models*, is to design a so-called *cost function* $\mathcal{J} : \mathbb{R}^m \to \mathbb{R}^+$. A cost function is a mathematical function, commonly convex, that quantifies the discrepancy between the observed data $\boldsymbol{y}$ and the reconstructed data $\boldsymbol{x}$, while simultaneously incorporating the prior knowledge about the scene. The cost function typically consists of two primary terms: A *data fidelity* term and a *regularization term*. The *data fidelity term* $\mathcal{F} : \mathbb{R}^m \to \mathbb{R}^+$ measures the consistency between the reconstructed image $\boldsymbol{x}$ and the observed data $\boldsymbol{y}$, usually using the *sensing model* $\boldsymbol{A}$. The *regularization term* $\mathcal{R} : \mathbb{R}^m \to \mathbb{R}^+$ typically encodes the prior knowledge and enforces the expected structures, spectral properties, or other physical characteristics of the scene. The total cost function $\mathcal{J}$ is the sum of those two terms:

$$\mathcal{J}(\boldsymbol{x}) = \mathcal{F}(\boldsymbol{x}) + \lambda \mathcal{R}(\boldsymbol{x}),$$

where $\lambda \in \mathbb{R}^+$ controls the balance between the *data fidelity term* and the *regularization term*.

The minimization of this cost function aims to find the optimal reconstruction that best aligns with the observed data, while following the constraints imposed by the prior knowledge:

$$\hat{\boldsymbol{x}} \in \arg\min_{\boldsymbol{x} \in \mathbb{R}^m} \mathcal{J}(\boldsymbol{x}). \tag{1}$$

The most common *data fidelity term* penalizes the discrepancy between the reconstruction candidate $\boldsymbol{x}$ and the observation vector $\boldsymbol{y}$ using the ℓ_2 norm:

$$\mathcal{F}(\boldsymbol{x}) = \|\boldsymbol{A}\boldsymbol{x} - \boldsymbol{y}\|_2^2.$$

In that case, the *data fidelity term* is convex and differentiable with respect to $\boldsymbol{x}$, making it a good candidate to minimize $\mathcal{J}$ through gradient descent. While convenient, this ℓ_2 norm is not the only option, and an ℓ_1-norm [55], [56] or a Kullback-Leibler divergence [57] are often used in the fidelity term.

On the other hand, the *regularization term* $\mathcal{R}$ highly depends on the problem to solve (deblurring, denoising, etc.) and the kind of prior knowledge on the data at hand. These priors are common in many RS tasks such as unmixing, deblurring, and fusion thanks to their ability to exploit the underlying *physics* of the observed scene.

The choice of prior is a critical design decision in fully model-based remote sensing methods, as it reflects both domain-specific knowledge and assumptions about the nature of the target signal or scene. By incorporating constraints such as spatial smoothness, spectral sparsity, or known physical bounds, the prior reduces ambiguity in otherwise ill-posed inverse problems and guides solutions toward physically plausible reconstructions. In practice, the effectiveness of a chosen prior is often revealed by how well it captures relevant structures without over-constraining the solution. For instance, total variation priors preserve sharp boundaries in land-cover classification, while sparse priors help recover distinct end-members in hyperspectral unmixing. Conversely, an ill-suited or overly restrictive prior can introduce systematic bias and prevent convergence. Thus, a well-designed prior not only reflects the underlying physics of the sensing problem but also strikes a balance between necessary regularization and sufficient flexibility to adapt to real-world data.

a) Total Variation (TV) Regularization: Minimizes the integral of the absolute gradient of an image or spectral profile, promoting piecewise-constant regions and sharper discontinuities. This makes TV especially useful for tasks like denoising and deblurring, where preserving edges (e.g., between roads and vegetation) is crucial. However, strong TV enforcement can introduce blur or a "staircasing" effect in smoothly varying regions if parameters are not balanced. TV is particularly useful in denoising (removing noise) [56], [58], unmixing [38], super-resolution [59], or pansharpening [60].

b) Sparse Regularization: Assume that the underlying signal is sparse or compressible in some representation (wavelets, dictionaries, learned transforms). Such priors are particularly effective in hyperspectral unmixing [61]–[64], super-resolution [65], [66], pansharpening [67], denoising [68], and other scenarios where high-dimensional data exhibit

TABLE I: Taxonomy of fully model-based, end-to-end, and hybrid methods.

Type of method	Strengths	Limitations
Fully model-based	• Fully interpretable (known physical models) • Predictable results and theoretical guarantees • Easily adaptable to new problems	• Needs accurate physical models • Many non-obvious design choices
End-to-End	• State-of-the-art performances • Learns complex physical models • Fast inference times once trained	• Not interpretable • Results may not follow physical laws • Requires expensive training
Hybrid	• State-of-the-art performances • Can account for physical model mismatches • Interpretable • Small neural networks	• Not easily adaptable to new problems • Training can be complex • Model-based theoretical guarantees may not apply

structured redundancy. Formally, such sparsity constraints are enforced by an ℓ_p norm (with $0 \leq p \leq 1$) on a desired representation of the data. By using such norms, these methods selectively encourage a few significant coefficients, preserving the main features while discarding noise or background redundancy.

c) Low-Rank Regularization: Uses the idea that multi-temporal or hyperspectral data can often be represented in a low-dimensional subspace. By restricting the data matrix (e.g., a stack of images or spectra) to be of low rank, one can denoise [69], [70] or complete missing observations with fewer degrees of freedom. This approach is widely used in applications like haze removal [37], demosaicking [71], and (pan)sharpening [19], [72]. Moreover, some approaches jointly combine sparse and low-rank regularization, such as the denoising approach presented in [73].

d) Non-Local Similarity: Exploits the fact that multi- and hyperspectral images contain repeated structures and textures across the scene. By searching a large spatial neighborhood for similar patches, one can group those patches and jointly process them to strongly suppress noise while preserving edges and fine details. One of the most common applications of non-local methods is denoising with the BM3D algorithm [74]. Still, the concept was quickly adapted to a wide range of problems, such as unmixing [63], deblurring [75], and image fusion [76].

It is common to observe overlaps between different methods, as they can often lead to comparable outcomes. For instance, sparse regularization frequently results in low-rank solutions, since sparsity constraints can be regarded as a relaxed form of low-rank modeling, as used in [13] for denoising. Moreover, [63] mixes sparse constraints with non-local similarity to solve unmixing problems; [58] combines total variation regularization with other sparsity constraints to tackle the strip noise removal problem.

Once the cost function $\mathcal{J}$ is fully defined with an observation $\boldsymbol{y}$ and a *sensing model* $\boldsymbol{A}$, the method used to minimize it is less critical. In general, there are no closed-form solutions to (1). Instead, many iterative algorithms have been proposed over the years to minimize cost functions $\mathcal{J}$, depending on the properties of the *data fidelity term* $\mathcal{F}$ and the *regularization term* $\mathcal{R}$. Those algorithms will start from an initial guess and successively apply the same operations until convergence or a stopping criterion is met.

For smooth and convex cost functions $\mathcal{J}$ (e.g., quadratic $\mathcal{F}$ and Tikhonov ℓ_2 regularization), gradient descent or conjugate gradient methods [85] can be used to converge to the minimum. When $\mathcal{R}$ is non-smooth but convex (such as an ℓ_1 or total variation prior), proximal gradient methods like the Iterative Shrinkage-Thresholding Algorithm (ISTA) and its accelerated variant Faster ISTA (FISTA) [86]–[88] are effective. More complex constraints can be handled via the family of Proximal Splitting Algorithms [89], such as the Alternating Direction Method of Multipliers (ADMM) ([37], [55], [57]), which splits the problem into subproblems that are easier to solve. Recent reviews [89] provide a comprehensive tour of such optimization algorithms for imaging.

Model-based methods have several well-known strengths. Since they derive from explicit physical and mathematical models, they are inherently interpretable: One can trace how each component of the model influences the reconstruction. They also tend to be robust to changes in input distributions as long as the models encoded by $\boldsymbol{A}$ and $\mathcal{R}$ remain valid, the method can be applied to new data without modification. Finally, model-based frameworks are flexible and generic, allowing the incorporation of new knowledge by modifying the cost function or simply adjusting the models. On the downside, designing a good model and prior can be labor-intensive and problem-specific. If the assumed models are inaccurate or oversimplified, the reconstructions will be biased or suboptimal. Moreover, minimizing $\mathcal{J}$ can be computationally costly, sometimes requiring hundreds of iterations for each image.

B. End-to-End Deep Learning Methods

The model-based and End-to-End deep learning approaches represent two distinct paradigms. In model-based methods, the primary focus lies on formulating a well-defined cost function that needs to be minimized to optimize the model's performance. In contrast, deep learning methods shift the focus from the explicit design of a cost function to the development of the learning algorithm itself. This emphasizes learning hierarchical representations of data, resulting in new cost functions.

In contrast to traditional methods that assume a predefined or known relationship between the ground truth and its corresponding observation, deep learning approaches aim to learn

TABLE II: Taxonomy of fully model-based methods. Note that those approaches are not mutually exclusive and can be combined to achieve better results.

Approaches	Strengths	Limitations	Applications
Total Variation (Review [79])	• Preserves sharp boundaries and significant edges • Effective for denoising, deblurring, or segmentation tasks • Convex	• Rare closed-form solution • Can over-smooth images • Not differentiable (ℓ_1 norm)	Denoising [58], [77], [78] Destriping [56], [58], [77], [80], Pansharpening [60], Super-resolution [59], Unmixing [38].
Sparse (Review [81])	• Helps to cast the problem in another domain (wavelets, etc.) • Discards noise and redundancy effectively • Convex	• Might take image content for artifacts • Not differentiable	Denoising [68], Super-resolution [65], [66], Unmixing [61], [63], [64], [78].
Low-Rank (Review [82])	• Exploits inherent low-dimensional structures • Strong data compression capacities	• Fails if data not actually low dimensional • Usually not convex • Complex rank estimation	Demosaicking [71], Denoising [78], Haze removal [37], Unmixing [83].
Non-Local (Review [74])	• Preserves edges and contours • Leads to low-rank structures	• Computationally expensive • Requires repetitive regions • Scales badly with images size	Deblurring [75], Denoising [74], Image Fusion [76], Super-resolution [84], Unmixing [63].

this relationship directly from data. Specifically, these methods learn a mapping from the observation space to the ground truth space by training on a dataset $\{\boldsymbol{x}_i, \boldsymbol{y}_i\}_{i=1}^N$ consisting of N pairs of ground truth values $\boldsymbol{x}_i \in \mathbb{R}^m$ and their corresponding observations $\boldsymbol{y}_i \in \mathbb{R}^n$. This approach not only enables the model to learn the direct relationship between observations and ground truth but also allows it to learn prior knowledge on the images themselves.

Formally, the goal is to train a neural network function $f_{\boldsymbol{\theta}} : \mathbb{R}^n \to \mathbb{R}^m$ with learnable parameters $\boldsymbol{\theta}$ being the solution of the following problem:

$$\hat{\boldsymbol{\theta}} \in \arg\min_{\boldsymbol{\theta}} \mathbb{E}_{(\boldsymbol{x},\boldsymbol{y})} \left[\mathcal{L}(\boldsymbol{x}, \boldsymbol{y}, f_{\boldsymbol{\theta}}) \right]. \tag{2}$$

As a result, the training process approximates (2) by calculating the average loss over the available dataset. In other words, instead of minimizing the expectation directly, the optimization routine minimizes the empirical mean of the loss computed across the training set:

$$\hat{\boldsymbol{\theta}} \in \arg\min_{\boldsymbol{\theta}} \frac{1}{N} \sum_{i=1}^{N} \mathcal{L}(\boldsymbol{x}_i, \boldsymbol{y}_i, f_{\boldsymbol{\theta}}). \tag{3}$$

A common choice of empirical training loss is the Mean Squared Error (MSE) loss $\frac{1}{N}\sum_{i=1}^N \|\boldsymbol{x}_i - f_{\boldsymbol{\theta}}(\boldsymbol{y}_i)\|_2^2$, which compares the ground truth and the reconstruction at a pixel scale.

However, in practice, the distribution of the images is unknown. Instead, we only have a finite dataset of N pairs examples $\{\boldsymbol{x}_i, \boldsymbol{y}_i\}_{i=1}^N$. Comparing (1) with (3) highlights the paradigm shift: in deep learning methods, one does not try to find the best reconstruction possible with a fixed algorithm, but instead tries to find the best algorithm that yields the best reconstruction in expectation.

Deep learning has proven remarkably powerful for RS image restoration and analysis, often outperforming classic methods in terms of fidelity and detail.

a) Convolutional Neural Networks (CNNs): CNNs were among the first architectures widely adopted in remote sensing ([126], [127]), thanks to their ability to model spatial context and hierarchical features. Such neural networks, like the widely used U-Net architecture [128], primarily employ convolutions with learned kernels to leverage the local correlation commonly found along the spatial dimensions of RS images. CNNs have been influential in tasks such as unmixing [99] and pansharpening [95].

b) Recurrent Neural Networks (RNNs): RNNs allow modeling temporal dependencies in remote sensing data by maintaining a hidden state that evolves through sequential inputs, allowing the network to capture dynamic patterns and contextual relationships across time. RNNs, and especially their variants such as Long Short-Term Memory (LSTM) [129], are able to extract temporal features for change detection [100], [101] or classification [103]–[105]. However, RNNs are less frequently used in image processing, as these tasks typically depend on spatial correlations, whereas RNNs are inherently designed to model temporal sequences.

c) Transformers: More recently, Transformer-based architectures have gained traction in remote sensing, following their success in computer vision [130]. They are based on self-attention, which weights the essential parts of the observed image and allows ViTs to capture correlations in data at a larger scale, in the spatial, temporal, and spectral dimensions. Vision Transformers (ViTs) architectures like the Shifted Window Transformer (Swin) [131] are thus able to grasp both spatial and temporal links in data [45], compared to CNNs and RNNs, which are specialized in either spatial correlations or temporal correlations. ViTs have been applied to a vast range of RS tasks, such as image deblurring [106], haze removal [109], super-resolution [110], and cloud removal [132].

d) Invertible Neural Networks (INNs): INNs are designed so that every transformation within the network is reversible. Unlike classical neural networks, INNs can learn

TABLE III: Taxonomy of deep learning methods. Note that those approaches are not mutually exclusive and can be combined to achieve better results.

Architecture	Strengths	Limitations	Applications
CNNs (Review [43])	• Strong spatial feature extraction • Well-suited for pixel-wise tasks like denoising, detection, and classification	• Emphasis on local regions • Relies on regular grid data (images)	Deblurring [90], [91], Denoising [92]–[94], Image fusion [21], Pansharpening [95], Super-resolution [93], [96]–[98], Unmixing [99].
RNNs (Review [102])	• Strong temporal feature extraction • Well-suited for multi-temporal data like vegetation monitoring	• Computationally expensive • Prone to vanishing gradient • Cannot capture spatial features	Change Detection [100], [101], Classification [103]–[105].
Transformers (Reviews [44], [45])	• Spatial and temporal extraction • Commonly pre-trained for fast fine-tuning • Effective in most remote sensing tasks	• Computationally expensive • Requires large datasets for pre-training	Deblurring [106], Denoising [107], [108], Haze removal [109], Super-resolution [110].
INNs (Review [112])	• Invertible mapping • Interpretable operations • Also learns physical models	• Restricted architecture design • Not robust to slight problem changes	Aerosol optical depth [111], Classification [113], Image compression [114], Pansharpening [115], [116].
Generative Models (Review [119])	• Synthesize realistic data • Learns spatial, spectral, and temporal trends	• Instable training and may collapse • Poor physical interpretability	Dehazing [117], [118], Pansharpening [120], [121], Super-resolution [122], [123], Unmixing [124], [125].

a bijective mapping from the observation distribution to a desired distribution (e.g., reconstructed images). Effectively learning how to process the observations, but also the physical models responsible for the observations. This property leads to better uncertainty quantification and interpretability. In remote sensing, INNs have a wide range of applications, especially when the known physical models are incomplete or too complex for practical use, as in aerosol optical depth [111], image compression [114], and pansharpening [115]. Moreover, the popular Normalizing Flows [133] techniques are a subset of INNs, where the observation distribution is simple (e.g. Gaussian distribution) and leverages probabilistic modeling.

e) Generative Models: Aim to learn the underlying data distribution of remote sensing imagery to synthesize, reconstruct, or enhance data by generating realistic samples consistent with observed patterns [134]. These models, whose main representatives are Generative Adversarial Networks (GANs), Variational Autoencoders (VAEs), and diffusion models, have become powerful tools for addressing data scarcity, resolution enhancement, and domain adaptation challenges in Earth observation. GAN-based approaches have been used for pansharpening [120], [121] and super-resolution [122] by learning to refine images through adversarial training between generator and discriminator networks [135]. VAEs [136], in contrast, model latent distributions explicitly and are often employed for spectral unmixing [124], [125]. More recently, diffusion probabilistic models [137] have shown strong generative capacity for high-resolution, encouraging physically consistent image synthesis, offering stable training compared to GANs and often exhibiting stronger generalization to diverse sensors [138]. They are successfully used in dehazing [117], [118], and super-resolution [123].

Similar to fully model-based methods, there is also overlap among deep learning strategies, aimed at combining the strengths of different methodologies. These overlaps primarily involve adapting a convolutional architecture to complement another approach. This arises from the fact that convolutions provide an effective mechanism for leveraging spatial correlations present in remote sensing images. For instance, [101] combines convolutional and recurrent neural networks to obtain a method able to grasp spatial, spectral, and temporal features in the context of change detection. Furthermore, [113] designs an invertible neural network based on a convolutional architecture to perform image classification; [125] uses a generative model based on convolutional networks to solve the unmixing problem.

Deep learning methods are now ubiquitous in remote sensing due to their ability to represent links in data that were previously beyond the capabilities of classical physical models. All those methods provide unprecedented performances in most RS tasks. On the other hand, they usually suffer from high computational cost, and their generalization capacity, along with their interpretability in physical terms, are still open challenges. Table III summarizes the strengths, limitations, and applications of the discussed deep learning methods.

IV. Hybrid Methods: Fusing Model-Based and Deep Learning Methods

Recently, significant efforts have been focused on integrating classical model-based methods with deep learning techniques [47], aiming to leverage the best aspects of both approaches. Classical model-based methods are valued for their interpretability, robustness, and reliance on physical knowledge. However, they may struggle with handling observations

deviating from the *sensing model* and perform poorly without correct prior assumptions.

On the other hand, deep learning methods have demonstrated strong performance, outperforming established model-based methods. However, deep learning models rely on large datasets and are often perceived as "black-box" techniques, lacking the interpretability and physical grounding that model-based methods provide.

Hybrid methods, sitting at the intersection of these two approaches, aim to combine the best of both worlds by integrating the interpretability and robustness of model-based methods with the performance of deep learning.

Deep Learning Assisted Methods can be seen as an extension of fully model-based methods. The *sensing models* are directly embedded into the methods, dictating the kind of operations that are performed. A key characteristic of these hybrid approaches is their focus on the cost function, just like fully model-based methods.

Furthermore, *Model-Guided Deep Learning Methods* follow the spirit of DL methods more closely. In such settings, the *sensing model* is only used to guide the learned network in its training phase or to create specific architectures that enforce desired properties by design.

Compared to end-to-end deep learning methods, hybrid methods generally require less training data. By constraining the solution space with physics (either through a sensing model or by limiting the architecture), the network has fewer "degrees of freedom" to learn, and is naturally guided toward a physically acceptable solution.

Compared to model-based methods, hybrid ones take full advantage of the recent learning capabilities and do not rely on arbitrary constraints. As a result, hybrid methods can be more data-efficient and tend to be more robust to setting changes or model mismatches compared to End-to-End methods, while achieving state-of-the-art results in many RS tasks.

A. Deep Learning Assisted Methods

In deep learning assisted methods, the cost function is designed similarly to how it is done in traditional model-based methods, as shown in (1), recalled here for convenience:

$$\hat{\boldsymbol{x}} \in \arg\min_{\boldsymbol{x}\in\mathbb{R}^m} \mathcal{F}(\boldsymbol{x}) + \mathcal{R}(\boldsymbol{x}).$$

However, unlike fully model-based methods where the *regularization term* is often manually chosen based on expert knowledge, hybrid methods offer more flexibility and adaptability.

One approach is to replace the hand-crafted *regularization term* by one induced by a pre-trained neural network. This approach enables deep learning models to influence the regularization process directly, learning the optimal form of regularization based on the available data. The key advantage of this method is that it removes the need for setting the *regularization term* in an ad hoc manner. This integration yields a more data-driven and adaptive solution to the optimization problem. Since deep learning-assisted methods are often designed to mirror their model-based counterparts closely, Fig. 4 presents the block diagrams of generic model-based and deep learning-assisted methods.

a) Plug-and-Play methods (PnP): PnP methods follow the same iterative framework as many fully model-based methods [169]. They break down the problem to solve into smaller subproblems, typically one related to the sensing model and one associated with the prior. However, compared to iterative model-based methods, PnP methods utilize a neural network instead of a handcrafted prior, often providing better constraints on the processed image than those manually defined by physical models. Their neural network is trained beforehand on a simple, controlled use case and is then plugged into the iterative scheme to take the sensing model into account and solve the desired problem. A key advantage of PnP is its modularity: The *physical model* (captured in the data fidelity term) and the learned *regularization term* (typically a denoiser) are treated independently. This implies that a learned regularization can be trained separately and reused across different tasks, imaging conditions, or sensors, without changing the flow of the entire algorithm.

Notably, [46] presents the advantages of turning two widely used model-based iterative algorithms, ADMM and ISTA, into plug-and-play methods across multiple applications. Furthermore, the paper discusses practical variants, such as online PnP, for enhancing scalability in large-scale imaging problems, which is especially beneficial in RS tasks involving large volumes of observed data.

Plug-and-Play methods have been successfully applied in remote sensing, particularly in areas such as hyperspectral unmixing, where Zhao et al. [146] derive a Plug-and-Play version of the ADMM algorithm [89] for recovering abundance maps under the assumption of known endmembers. Additionally, PnP methods are used in pansharpening, such as [144] where ADMM PnP is employed to minimize a cost function comprising three terms: A data fidelity term, and a CNN-based prior, by the typical structure of PnP approaches, but with a third term taking into account spatial consistency using total variation.

Fig. 4 presents the Plug-and-Play variant of a model-based algorithm. In this formulation, the explicit *regularization term* $\mathcal{R}$ is replaced by a pretrained neural network $N_{\boldsymbol{\theta}}$, typically a CNN. This substitution is motivated by the fact that the corresponding step in the original method effectively solves a classical denoising problem, making a denoiser a natural choice for this role [46].

Recently, diffusion models [137], [170] have been used as denoisers within the PnP framework [171]–[173], replacing classical CNNs in traditional image processing tasks. The iterative structure of diffusion models aligns naturally with the iterative framework of PnP methods, enabling more precisely adapted operations at each step and even supporting multimodal integration by incorporating textual information alongside the observations [172].

Employing a learned denoiser as a regularizer does not invalidate all theoretical guarantees. Under some conditions on the learned regularization, the resulting PnP method provides similar convergence guarantees to their fully model-based

TABLE IV: Taxonomy of hybrid methods.

Architecture	Strengths	Limitations	Applications
PnP (Review [46])	• No need for hand-crafted regularizations • No need for large neural networks • One learned prior can be used for multiple problems • Retains convergence guarantees	• Depends on the accuracy of the sensing model • Remains tuneable parameters	Deconvolution [139] Denoising [140]–[142], Image compression [141], [143], Pansharpening [144], Sharpening [145], Super-resolution [141], Unmixing [146].
Unrolling (Review [48])	• Great learning capacities while retaining interpretability • No parameters to tune by hand • Usually faster than its model-based counterpart	• Overfits on the problem to solve • Difficult to train, prone to vanishing gradients	Denoising [147], Image compression [148], [149], Pansharpening [150], Sharpening [151]–[153], Unmixing [154], [155].
DIP (Review [49])	• No need for hand-crafted regularizations • No preliminary training • Easily adaptable to different tasks	• Computationally expensive • Prior not easily interpretable • Needs appropriate early stopping	Denoising [156], Pansharpening [157], Sharpening [152], [153], [158], Super-resolution [159]–[161], Unmixing [162].
Model-Inspired Architectures	• Full consistency with physics • Strong interpretability	• Tailored for specific applications • Reduced learning capacity	Deblurring [163], Dehazing [164], [165], Sharpening [166], Unmixing [167], [168].

counterparts [174]. Most notably, fixed-point convergence can be proven [175] in the case of ADMM PnP, and practical instructions are given by [176] to satisfy the convergence conditions.

b) Unrolling (or Unfolding): While PnP methods have proven successful in RS image processing by removing the need for regularization tuning, they still contain tunable parameters and do not fully use recent deep learning capacities. A step further in the deep learning direction leads to unrolled methods. Instead of just learning the *regularization term*, algorithm unrolling proposes to unroll a fixed number of iterations as a neural network [48]. Its architecture replicates the structure of the original iterative algorithm by explicitly modeling each iteration as a corresponding neural network block, resulting in a repeated and interpretable design. The algorithm's variables at each iteration become network layers, and parameters (such as step sizes or threshold values) that were previously fixed or manually tuned are converted into learnable weights. Moreover, the learned parameters and the *regularization term* are not constrained to be the same at each iteration, allowing more learning capacity. Unlike Plug-and-Play, which uses a fixed regularization module, unrolling learns a whole sequence of operations using a predefined *sensing model*, allowing the network to adapt differently at each iteration. However, this often leads to an overly specialized method on a specific problem, and changing the *sensing model* is not enough to adapt it to newer configurations.

Fig. 4 illustrates the modifications required to transform a fully model-based method into its unrolled counterpart. The central change involves allowing the network's parameters $\boldsymbol{\theta}$ to vary across iterations, enabling the regularizer to learn iteration-specific operations. However, this design imposes a limitation: The number of iterations must be predetermined, as each iteration corresponds to a distinct set of learned parameters that must be trained in advance.

Unrolling methods are not truly iterative, as each iteration differs from the others, and the algorithm cannot be executed for more iterations than those specified during training. This significantly complicates theoretical exploration of the method, as no convergence guarantees can be easily defined. However, some progress has been made in this direction [177], introducing stronger constraints on the learned regularizer than those used in PnP, while remaining feasible for practical applications.

For example, algorithm unrolling is employed in blind unmixing [155], where both the abundance maps and the endmembers are estimated using nonnegative matrix factorization. Moreover, unrolling has also proven successful in sharpening [152], [153], where the resulting unrolled algorithm is used without preliminary training (see Section IV-B) to sharpen Sentinel-2 images.

Finally, unrolling was also applied to the demosaicking problem [178], where the authors address demosaicking under different configurations in a unified way. As shown in Fig. 5, ADMM produces reasonable outputs but suffers from artifacts. In contrast, fully data-driven methods [179] trained under specific configurations struggle to generalize to newer ones. On the other hand, the unrolled method, while trained on specific configurations, is flexible enough for newer configurations, effectively merging the flexibility of fully model-based methods with the performance of deep learning methods. Quantitatively, this study demonstrates that the proposed ADMM unrolled method surpasses its classical ADMM counterpart by an average margin of 7 dB in PSNR (Peak Signal-to-Noise Ratio), while also delivering consistent performance across a broad range of modalities compared to fully end-to-end approaches.

B. Model-Guided Deep Learning Methods

Model-guided deep learning methods refer to approaches in which the *sensing model* plays a central role in guiding the learning process or the design of neural networks. Two primary strategies can be identified within this paradigm. The first involves conventional deep learning techniques that incorporate the *sensing model* to learn informative representations. Representative examples include Deep Image Prior (DIP) [49]

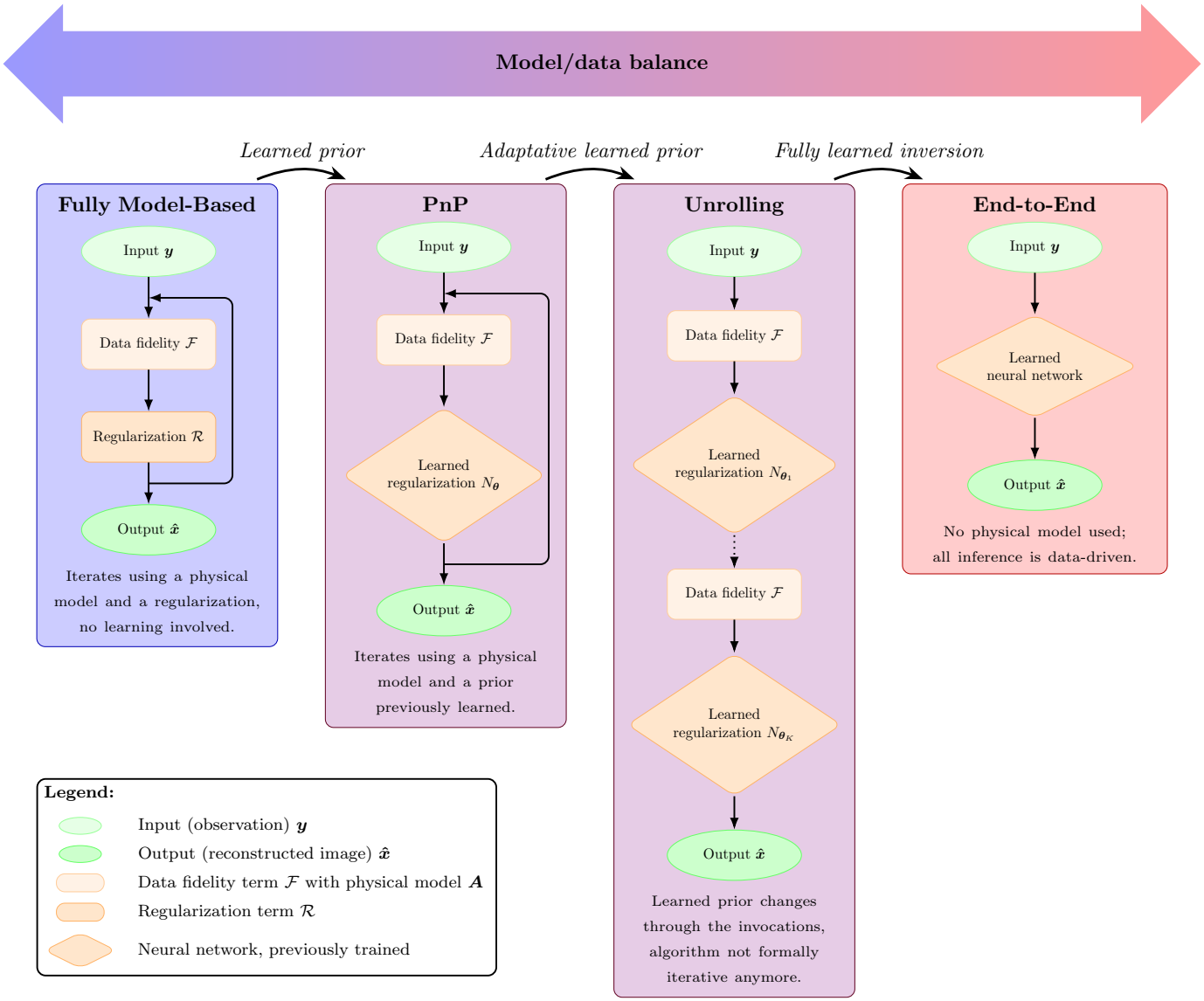


Fig. 4: Flowchart of the different deep learning assisted methods. This branch of hybrid HDL methods follows the philosophy of MB ones, borrowing the iterative backbone. PnP stays an iterative method where the regularization term is learned. Unrolling methods break this iterative scheme by changing the learned regularization at each iteration.

and other unsupervised learning frameworks. These methods typically formulate custom loss functions that explicitly embed the *sensing model* into the training process.

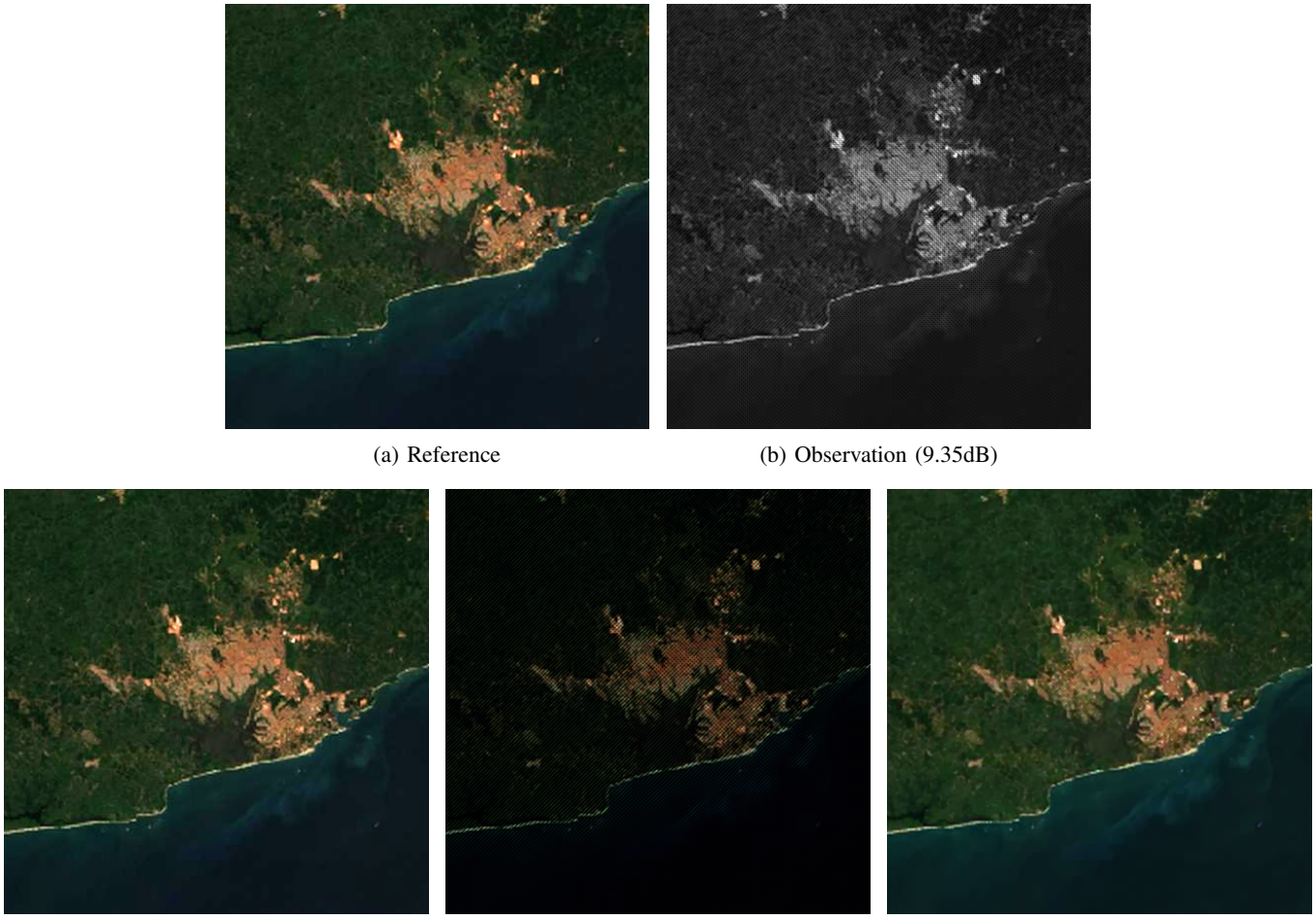
The second strategy corresponds to model-inspired architectures, where the architecture of the neural network is implicitly designed by knowledge encapsulated in the *sensing model*. This approach, leaning toward classical DL methods, aims to ensure that the resulting architecture adheres to specific, desirable properties grounded in physical or mathematical principles.

a) Deep Image Prior (DIP): The term "unsupervised learning" encompasses a wide range of definitions within the context of RS image reconstruction. Among the various approaches, the Deep Image Prior (DIP) framework is particularly relevant for HDL methods, as it exemplifies the integration of prior knowledge through the network architecture without requiring external supervision.

DIP is an iterative, training-free, and unsupervised method, similar to fully model-based methods. Unlike supervised DL methods that require large annotated datasets, DIP operates solely on the observation and leverages the in-built assumptions of convolutional neural networks (CNNs). In particular, the network architecture itself acts as a regularization that favors natural image statistics, enabling high-quality reconstructions even in the absence of external training data.

As illustrated in Fig. 6, the DIP framework involves optimizing a neural network N (typically a CNN) with parameters $\boldsymbol{\theta}$. The objective is to minimize a data fidelity term $\mathcal{F}$ of the form $\|\boldsymbol{A}\boldsymbol{x} - \boldsymbol{y}\|_2^2$, where $\boldsymbol{x}$ is the network's output. Here, the regularization is not imposed explicitly but emerges from the structure of the CNN, which inherently promotes spatial coherence and smoothness due to the convolutional filters.

Typically, the network's input is fixed random noise $\boldsymbol{\epsilon}$, and



(a) Reference (b) Observation (9.35dB)

(c) Unrolling (42.35dB) [178] (d) Fully end-to-end deep learning (17.63dB) [179] (e) ADMM (34.39dB) [89]

Fig. 5: Demosaicking results on simulated RGB Sentinel-2 observations and PSNR value in dB. The unrolling method [178] is flexible to out of distribution configurations, while the end-to-end method [179] does not. Conversely, classical ADMM [89] works but produces significant artifacts. The full end-to-end deep learning method fails to reconstruct observations taken in a different scenario than seen during training. The fully model-based method gives a blurred result. Conversely, the unrolled method produces a sharper image without visible artifacts.

the parameters $\boldsymbol{\theta}$ are optimized so that the output matches the observation through a data fidelity term. In this setting, one step in optimizing the parameters $\boldsymbol{\theta}$ is analogous to one iteration of a fully model-based method. Despite its simplicity, DIP has proven effective for a range of inverse problems such as sharpening [161], where the authors focus on designing a neural network specially tailored for image fusion by generating coarse estimations from the observations and using them as input to a neural network similar to U-Net. DIP is also capable of addressing the blind unmixing problem [162], where the endmembers are first estimated based on the geometry of the observations; DIP is then used to obtain abundance maps given the estimated endmembers.

Because DIP essentially consists of a neural network to be trained, its theoretical guarantees are more limited than those of PnP or unrolling. In DIP, training typically reduces the data-fidelity term $\mathcal{F}$, but this does not ensure convergence to a physically meaningful solution. The prior is implicit and depends on the network architecture, initialization, and the optimization process (e.g., early stopping), so minimizing $\mathcal{F}$ alone may still yield artifacts or overfitting.

Despite the conceptual simplicity of DIP, the learned neural network may still be prone to overfitting the observed data [49], as the implicit regularization imposed by the network architecture alone may not sufficiently constrain the solution toward an accurate reconstruction [49]. To mitigate this issue, it is necessary to determine, empirically or automatically, a suitable early stopping strategy to halt the optimization process before overfitting occurs. Moreover, as shown by [158] in the case of Sentinel-2 and Landsat 8 super-resolution, the DIP framework can be further improved by adding an explicit *regularization term* that helps to reconstruct a low-rank solution. The proposed unsupervised hybrid method sharpens images with an initial spatial resolution of 60, 30, 20, or 15 meters to a target of 10 meters, delivering clear and consistent details. The implicit regularization introduced by the neural

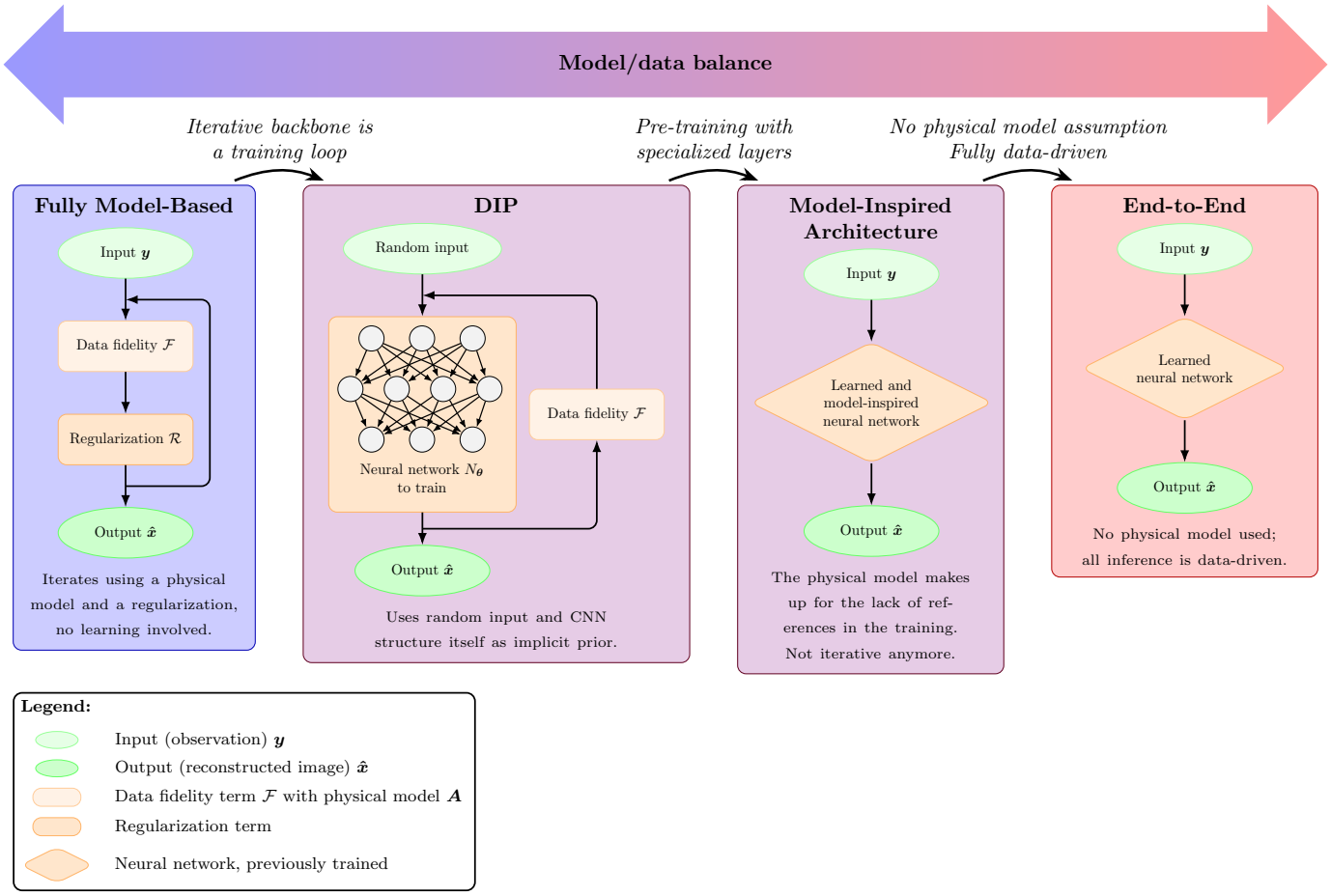


Fig. 6: Flowchart of the different model-guided methods. This branch of HDL methods follows the philosophy of DL ones, where the neural network occupies a central part of the method. DIP uses the DL framework as an iterative method, similar to a model-based one. Model-inspired architectures goes further away from model-based methods by designing task-specific architectures.

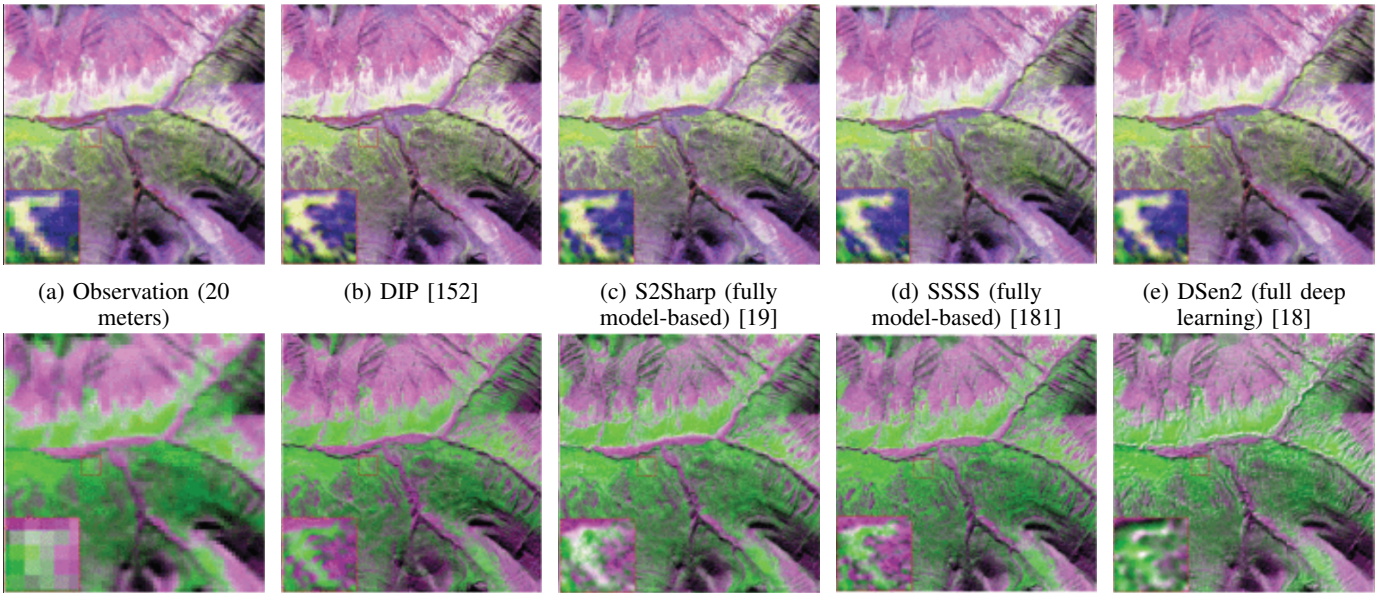
network gives a superior visual quality compared to fully model-based methods.

Additionally, in the context of Sentinel-2 single image super-resolution, [152], [153] show that the DIP framework can be expanded by leveraging Stein's Unbiased Risk Estimator (SURE) [180] in the loss function to reduce the risk of overfitting. This extension is shown to enhance reconstruction quality, with the proposed method outperforming both model-based and deep learning approaches by an average of 4 dB in SRE (Signal-to-Reconstruction Error) across multiple noise scenarios. As shown in Fig. 7, the proposed hybrid method effectively consistently enhances the details of the observations compared to fully model-based methods ([19], [181]) and does not produce blurred results like the full deep learning method DSen2 [18].

b) Model-Inspired Architecture: An alternative to incorporating the *sensing model* within the loss function is to utilize it as a foundation for the architectural design of neural networks. In this approach, prior knowledge of the sensing process is used to design network architectures that intrinsically enforce physically meaningful constraints or adhere to specific properties of the underlying data generation model.

Model-inspired methods can offer strict guarantees that the outputs adhere to the desired properties; Since the architecture is often explicitly designed to enforce the wanted constraints, the training process cannot deviate from them. However, this often rigid design may overly restrict the model, potentially limiting its expressiveness and its ability to accurately reconstruct the target images.

These methods are typically designed for specific RS tasks, as their architectures are inherently problem-dependent. For instance, in the context of blind hyperspectral unmixing, it is common to impose physical constraints such as the non-negativity of abundance values and the requirement that they sum to one per pixel. These constraints can be incorporated directly into the network architecture through specialized layers that enforce them by design, as demonstrated in [167]. The benefits of such a model-inspired architecture over a traditional model-based method are illustrated in Fig. 8, which presents a qualitative evaluation of unmixing performance on the Samson dataset for the "Soil," "Tree," and "Water" endmembers. The proposed model-inspired architecture consistently outperformed the compared model-based approaches, reducing the SAD (Spectral Angle Distance) by an order of



(a) Observation (20 meters) (b) DIP [152] (c) S2Sharp (fully model-based) [19] (d) SSSS (fully model-based) [181] (e) DSen2 (full deep learning) [18]

(f) Observation (60 meters) (g) DIP [152] (h) S2Sharp (fully model-based) [19] (i) SSSS (fully model-based) [181] (j) DSen2 (full deep learning) [18]

Fig. 7: False color representation of the super-resolution to 10 meters of 20 meters (top row) and 60 meters (bottom row) Sentinel-2 real observations [152]. While all methods give significantly sharper reconstructed images than the observed images, S2Sharp creates artifacts, SSSS, and DSen2 yield 60 meters images that are blurrier than one yielded by DIP.

magnitude in the unmixing experiments.

Similar strategies have been applied to image dehazing [164], [165], where the designed network mimics the atmospheric scattering model to ensure the output is consistent with the underlying physical phenomena occurring. Moreover, [163] solves the hyperspectral deblurring problem by estimating the blurring kernel and performing Wiener deconvolution in the feature space.

C. Choosing the Right Hybrid Method

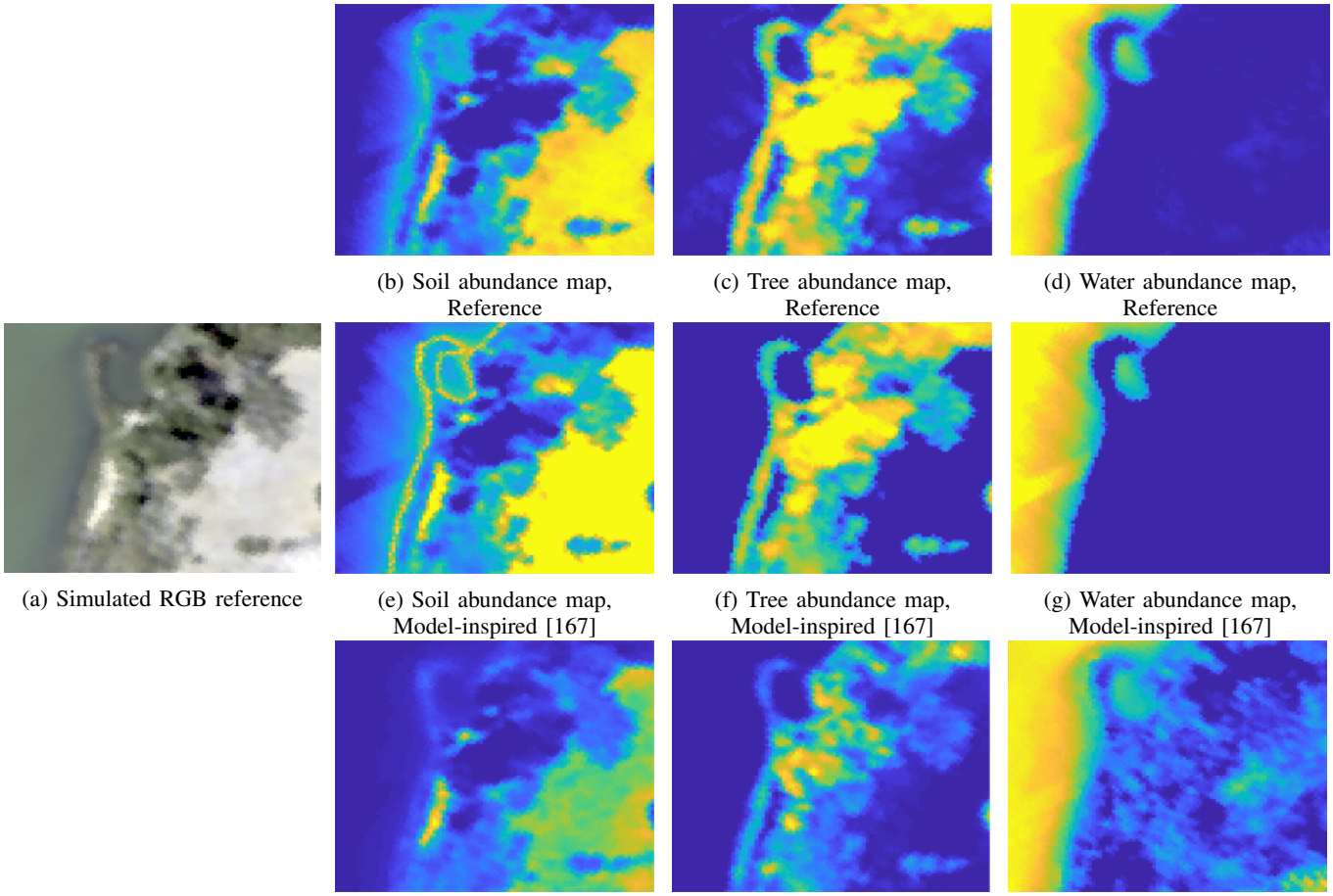
Selecting the appropriate hybrid method is particularly important in RS image processing tasks, where the goal is often to reconstruct high-fidelity spatial or spectral information from limited or noisy measurements, usually coming from a wide variety of sensors. Choosing the appropriate hybrid method depends on the desired level of physical correctness, as it determines the extent to which the reconstruction adheres to the underlying *physical model* and respects the inherent physical constraints of the world. In this context, hybrid methods must balance the integration of physical prior, such as *sensing models*, with the adaptability of neural networks to complex, real-world variations. Fig. 9 provides a visual guide to help identify which class of hybrid techniques is best suited for a given scenario, depending on the available prior knowledge, data constraints, and the importance of adhering to physical models.

One of the most critical criteria for selecting among various hybrid deep learning methods is the requirement for physical correctness. PnP methods, for instance, provide interesting convergence guarantees under certain assumptions regarding the learned denoiser [174], thus promoting a theoretically sound asymptotic reconstruction. Additionally, model-guided architectures inherently promote physical consistency by restricting the network's operations to predefined, physically meaningful transformations. Conversely, methods such as unrolling and DIP do not offer comparable theoretical guarantees, but they allow for greater flexibility in the reconstruction process due to fewer structural constraints.

While all hybrid deep learning methods require prior knowledge of the acquisition process, typically in the form of a *sensing model*, not all need access to additional training data. Such methods, like PnP (when used with a pre-existing denoiser) or DIP, are useful when not enough data are available for training. Furthermore, the absence of a training-inference mismatch in these methods enhances their adaptability to novel problems or data sources.

In addition to the previously discussed criteria concerning physical meaningfulness and training aspects, it is crucial to select a method that aligns with the problem's dimensionality and the available computational resources. Indeed, unrolling methods benefit from a high learning capacity, as the regularization is not constrained to remain fixed across iterations. However, this strength may imply numerous trainable parameters, as the parameter count increases linearly with the number of iterations. In contrast, PnP methods avoid this issue, since the learned regularization is shared across all iterations, making the parameter count constant.

Moreover, DIP methods tend to be relatively slow when processing large remote sensing images. This limitation arises from the need to train a complete neural network from scratch for each image over numerous iterations. As a result, DIP methods can require computation times exceeding one minute, making them unsuitable for time-critical or real-time



(a) Simulated RGB reference

(b) Soil abundance map, Reference

(c) Tree abundance map, Reference

(d) Water abundance map, Reference

(e) Soil abundance map, Model-inspired [167]

(f) Tree abundance map, Model-inspired [167]

(g) Water abundance map, Model-inspired [167]

(h) Soil abundance map, Model-based [182]

(i) Tree abundance map, Model-based [182]

(j) Water abundance map, Model-based [182]

Fig. 8: Estimated abundance maps for the Samson dataset of the endmembers "Soil", "Tree", and "Water". First row is the reference, second row is the proposed hybrid method [167], and third row is a model-based method [182]. The maps estimated by the model-inspired method are much closer to the reference (SAD of 0.031 from reference) compared to the fully model-based method (SAD of 0.0472 from reference).

applications.

Finally, model-inspired architectures are typically tailored for a specific task, which makes them particularly computationally efficient once training is complete. However, unlike other hybrid methods, model-inspired architectures are not well-suited for addressing multiple image processing tasks simultaneously, as a new model must be designed and trained for each application. While this limitation poses little concern for single-task scenarios, handling multiple tasks, such as joint denoising and deblurring, significantly increases the overall pipeline complexity.

V. Conclusion and Research Perspectives

In this paper, we presented a comprehensive overview of hybrid deep learning methodologies developed for remote sensing processing tasks. These methods strategically integrate recent advances in deep learning with well-established physical models, aiming to harness the complementary advantages of both paradigms. By bridging data-driven learning with domain-specific physical knowledge, the surveyed methods demonstrate enhanced performance, improved generalization across sensing conditions, and increased interpretability compared to their purely model-based and deep learning counterparts, thus offering promising directions for the next generation of RS solutions.

However, several challenges persist and represent interesting research directions. For example, a notable gap exists between the fields of image processing and remote sensing regarding the development and integration of hybrid methodologies. Indeed, the image processing field has a flourishing literature on hybrid methods, notably the use of recent diffusion models [137], [170] as denoisers within the PnP framework [171], [173], which have not been adopted in RS tasks. Adapting existing hybrid methods from the image processing domain to RS observations is not a straightforward task, as RS data exhibit fundamentally different characteristics compared to conventional natural images. Unlike standard RGB images, RS observations often involve multi- or hyperspectral data with high spectral resolution, varying spatial resolution across the spectral bands, temporal coverage constrained by satellite revisit intervals, and diverse acquisition geometries, including aerial, satellite, and drone-based platforms. Consequently, the

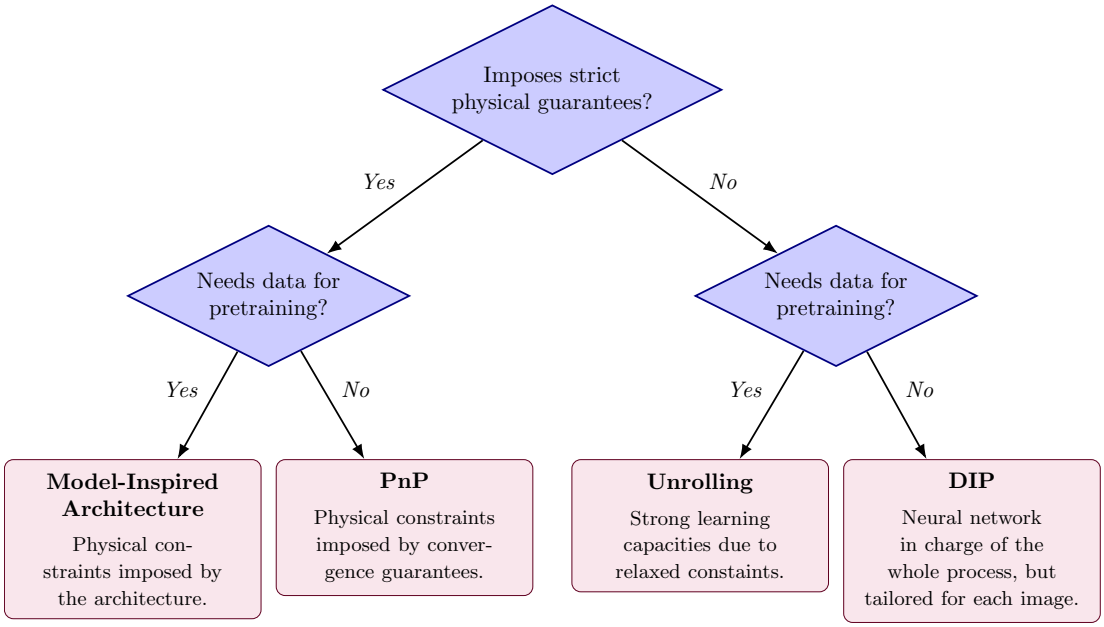


Fig. 9: Decision diagram for selecting the appropriate hybrid method based on the required level of physical correctness and the need for training data.

successful transfer of hybrid methods to the RS domain requires careful consideration of these distinctive features.

Additionally, the development of foundation models (FMs) has gained increasing interest in remote sensing [183]–[185]. Recent research explores the integration of multiple image modalities to obtain richer representations that perform well on various downstream tasks (e.g., image classification, change detection). However, to our knowledge, existing FMs are not yet trained using hybrid approaches (and thus lack physics-based information content). We believe that FMs that consider hybrid architectures, which can then support the downstream tasks associated with RS image processing (e.g., compression, unmixing) are needed. Inspired by the success of large language models (LLMs)-based FMs, such as Llama and GPT-3, vision-language models (VLMs) in RS have recently exhibited substantial advancements. Existing VLMs in RS demonstrate remarkable capability in several image-language tasks, e.g., image-text retrieval, visual question answering and image captioning. These developments pave the path for the design of LLM-powered agent frameworks. We believe that emerging AI agent technologies offer promising solutions for automating complex workflows related to remote sensing image processing tasks, such as autonomously managing remote sensing data, selecting tools, and refining outputs for these tasks.

Another promising direction for future research lies in the theoretical investigation of hybrid approaches that combine model-based and deep learning methods. While purely model-based methods have been extensively studied in terms of their mathematical foundations, convergence properties, and physical interpretability, hybrid methods often lack comparable theoretical grounding. For instance, unrolled methods, despite being inspired by iterative model-based algorithms, frequently lose the formal guarantees (such as convergence and stability) that their traditional counterparts offer. Similarly, DIP guides the reconstruction process using a physically meaningful loss function; However, they do not inherently ensure that the final output adheres to underlying physical constraints or yields a physically plausible solution. This absence of formal grounding underscores a critical need for more comprehensive analytical frameworks that can bridge the interpretability and reliability gap between conventional and hybrid paradigms.

Finally, the majority of hybrid methods employ deep learning primarily as a form of prior constraint to guide the reconstruction of the target images. These priors typically capture intrinsic properties of RS observations, such as texture, structure, or smoothness, rather than characteristics specific to the application or sensing modality. However, the employed neural networks are frequently trained in a task-specific manner, thus limiting their generalizability across different inverse problems or domains. This limitation points to the need for a more general solution. A compelling direction for future research involves the development of deep learning architectures capable of acting as "universal priors"; That is, models that encapsulate general image statistics in a manner independent of the processing task. This concept aligns with the emerging paradigm of foundation models, which aim to learn rich, transferable representations that can be adapted to a wide range of image reconstruction and analysis tasks with minimal task-specific tuning. The notion of universality is particularly relevant in remote sensing, where data acquired from diverse sensors and platforms often share common structural properties, such as spatial coherence, spectral redundancy, and multi-scale object patterns. A well-designed universal prior could effectively capture these shared characteristics, enabling more robust and adaptable hybrid methods.

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